MBARARA UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF COMPUTING AND INFORMATICS
COMPUTER SCIENCE

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Reg no: 2025/BCS/084/PS
Course: Database Systems
Course code: CSC1204
Lecturer: Mr. Baguma Kenneth
Assignment: Database Project 1
Start Date: 17th March 2026
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Discussion Group: 12

PROJECT REPORT

GROUP CHOSEN AREA OF INTEREST: AGRICULTURE

PROJECT TITLE: Farm Machinery and Equipment Management Database System

2025/BCS/084/PS
KABUNGA CHARLES
Group: 12

DATABASE PROJECT

Agriculture – Farm Machinery and Equipment Management

This area focuses on managing;

- Tractors, ploughs, harvesters, basic farm tools among others.
- Equipment usage schedules
- Maintenance records
- Fuel consumption
- Operator assignments

Chosen System: Farm Machinery and Equipment Management Database System in the agriculture field

Mission of the project

To design and implement a reliable, efficient, and scalable database system that facilitates effective management, monitoring, and analysis of agricultural machinery and farm equipment to improve productivity and decision-making.

General Objective

To develop a centralized database system for managing agricultural machinery and farm equipment.

Specific Objectives

- To store and organize detailed information about farm machinery and equipment
- To track equipment availability, allocation, and operational status
- To manage maintenance schedules and repair records
- To control user access and system security
- To improve efficiency in data retrieval and management
- To generate reports for informed decision-making
- To monitor costs associated with equipment usage and maintenance

- To ensure data integrity and consistency

Reasons:

Machinery is central to modern agriculture productivity therefore highly relevant in the agricultural field

The broad scope of machinery combines inventory, maintenance, and usage tracking of farm tools hence highly used.

The database system addresses problems like;

- ✓ Equipment misuse
- ✓ Poor maintenance tracking thus leeds to the loss of machinery
- ✓ Farms can later integrate with advanced tools and records

Functions of the database system

- Registering all farm equipment
- Tracking equipment usage (who used it, when and for what task)
- Scheduling maintenance and repairs
- Monitoring availability and allocation
- Recording fuel consumption
- Generating reports for management

User Groups

- Farm managers
- Equipment operators
- Maintenance technicians
- Agricultural officers / supervisors

System Benefits after implementation

- Systematic reduced maintenance costs
- Increased equipment lifespan
- Improved utilization of machinery

- Reduced fuel wastage
- Better planning of farm operations
- Improved decision-making through data insights
- Increased accountability of workers
- Better record keeping and transparency
- Enhanced productivity and efficiency
- Scalablity is maximised thus easy retrieval of records

Analysis of Costs

- Tangible Costs:
- Hardware (computers, servers)
- Software development
- Training costs
- Maintenance and updates

Expected hardships

- Resistance to change from staff
- Time spent learning the system
- Temporary productivity drop during transition

Benefits outweigh costs, especially long-term (cost savings + efficiency gains)

User Issues

- Skill Levels:
- Some users may have limited ICT skills
- Training Needs:
- Basic computer and system usage training required
- Adoption Challenges:
- Resistance from workers used to manual systems

Mitigation to the user issues

- Simple User Interface thus easy tov adapt
- Conduct training sessions
- Offer user manuals and support

Risk Assessment

Technical Risks	Mitigation
➢ System failure or bugs ➢ Data loss or corruption ➢ Poor system design	➢ Regular backups ➢ Testing before deployment

Organizational Risks	Mitigation
➢ Resistance to adoption ➢ Lack of management support	➢ Stakeholder involvement ➢ Awareness and training

Operational Risks	Mitigation
➢ Incorrect data entry ➢ Misuse of system	➢ Validation rules ➢ User access control

Limitations of the System

May not support real-time tracking without IoT integration

Depends on user input accuracy

Limited functionality in areas with poor electricity or internet

Initial setup cost may be high for small-scale farmers

Does not fully automate farm operations (only supports management)

PROJECT ENTITIES AND THEIR ATTRIBUTES

1. **Equipment**

equipment_id (PK)

equipment_name

category_id (FK)

category_name

status

2. Branch

branch_id(PK)

manager

branch_name

3. **User**

user_id (PK)

name

email

phone

qualification

location

role_id

role_name

4. **Task**

task_id (PK)

task_name

5. Deployment

deployment_id (PK)

equipment_id (FK)

user_id (FK)

start_date

end_date

task_name

6. **Maintenance**

maintenance_id (PK)

equipment_id (FK)

user_id (FK)

date

description

amount

7. **FuelLog**

fuel_id (PK)

equipment_id (FK)

date

quantity

cost

userName(FK)

8. **Expense**

expense_id (PK)

equipment_id (FK)

expense_category

amount

date

Binary Relationships between entities

Equipment → Branch (Many-to-One)

Equipment → User (One-to-Many)

Equipment → Maintenance (zero-to-many)

Equipment → Deployment (zero-to-many)

Equipment → FuelLog (One-to-Many)

User → equipment (zero-to-many)

User → Task (Many-to-One)

User → Deployment (zero-to-many)

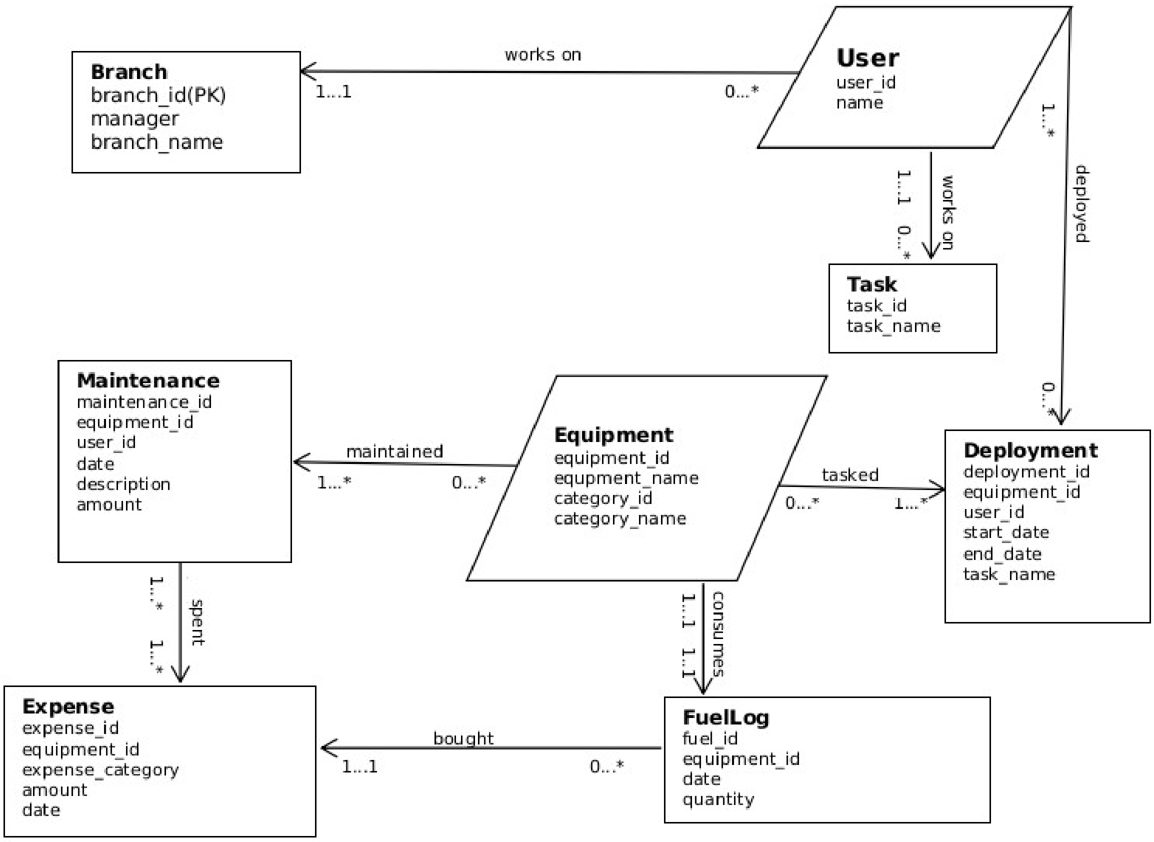
User → Maintenance (zero-to-many)

FuelLog → Expense (one-to-one)

User → Branch (one-to-one)

Maintainence → Expense (one-to-many)

ENTITY RELATIONAL DIAGRAM



THE PHYSICAL SCHEMA

I created the database and named it FarmMachineryAndEquipmentManagement

```
mysql> SHOW DATABASES;
+-------------------------------------+
| Database                            |
+-------------------------------------+
| BCS2025                             |
| FarmMachineryAndEquipmentManagement |
| bcs2025                             |
| information_schema                  |
| mysql                               |
| performance_schema                  |
| sys                                 |
+-------------------------------------+
7 rows in set (0.00 sec)
```

I inserted the tables into the database and their Attributes.

Using the command,

INSERT TABLE tableName(Attribute1,attribute2,....);

I inserted the records into the attributes using commands like;

```
mysql> INSERT INTO user VALUES("U0001", "Kwesiga Devine", "kwesigadevine001@gmail.com", 0702378436, "Diploma mechanics", "Masaka", "R001", "Manager");
Query OK, 1 row affected (0.18 sec)

mysql> INSERT INTO user VALUES("U0002", "Tusiime Maria Desire", "tusiimemariad001@gmail.com", 0776332126, "BSc Agriculture", "Mbarara", "R003", "Superviser");
Query OK, 1 row affected (0.43 sec)

mysql> INSERT INTO user VALUES("U0002", "Tusiime Maria Desire", "tusiimemariad001@gmail.com", 0776332126, "BSc Agriculture", "Mbarara", "R003", "Superviser");
Query OK, 1 row affected (0.14 sec)

mysql> INSERT INTO user VALUES("U0005", "Kabunga Charles", "kabungacharles001@gmail.com", 0727642126, "MSc Engineering", "Masaka", "R005", "Operator");
Query OK, 1 row affected (0.48 sec)

mysql> INSERT INTO user VALUES("U0004", "Mubbiru Clever", "mubbiruclever@gmail.com", 0754674352, "Diploma Agronomy", "Entebbe", "R004", "Technician");
Query OK, 1 row affected (0.44 sec)
```

After successful INSERTION of the tables in my database,

```
mysql> SHOW TABLES;
+---------------------------------------------+
| Tables_in_FarmMachineryAndEquipmentManagement |
+---------------------------------------------+
| branch                                      |
| deployment                                  |
| equipment                                   |
| expense                                     |
| fuelLog                                     |
| maintenance                                 |
| task                                        |
| user                                        |
+---------------------------------------------+
8 rows in set (0.01 sec)
```

After inserting values in the attributes of the database,

I queried to see the table content using the **SELECT** command.

Task entity

```
mysql> SELECT * FROM task;
+--------+---------------+
| taskId | taskName      |
+--------+---------------+
|      1 | Cultivation   |
|      2 | Maintenance   |
|      3 | Weeding       |
|      4 | Spraying      |
|      5 | Destructuring |
+--------+---------------+
5 rows in set (0.00 sec)
```

User entity

```
mysql> SELECT * FROM user;
+--------+----------------------+----------------------------+-----------+-------------------+----------+--------+------------+
| userId | name                 | email                      | phone     | qualification     | location | roleId | roleName   |
+--------+----------------------+----------------------------+-----------+-------------------+----------+--------+------------+
| U0001  | Kwesiga Devine       | kwesigadevine001@gmail.com | 702378436 | Diploma mechanics | Masaka   | R001   | Manager    |
| U0005  | Kabunga Charles      | kabungacharles001@gmail.com | 727642126 | MSc Engineering   | Masaka   | R005   | Operator   |
| U0004  | Mubbiru Clever       | mubbiruclever@gmail.com    | 754674352 | Diploma Agronomy  | Entebbe  | R004   | Technician |
| U0003  | Tumwiine David       | tumwiinedavid111@gmail.com | 743563487 | BSc Farming       | Kamwokya | R002   | Admin      |
| U0002  | Tusiime Maria Desire | tusiimemariad001@gmail.com | 776332126 | BSc Agriculture   | Mbarara  | R003   | Superviser |
+--------+----------------------+----------------------------+-----------+-------------------+----------+--------+------------+
5 rows in set (0.00 sec)
```

Maintenance entity

mysql> SELECT * FROM maintenance;

maintenanceId	equipmentId	userId	date	description	amount	endDate	taskName
M0001	EQ002	U0001	20-03-2026	Welding rod replace	150000	21-03-2026	Replacement
M0002	EQ004	U0002	24-03-2026	Blade shappening	150000	25-03-2026	Sharpening
M0003	EQ005	EQ004	22-03-2026	Drill bit change	110000	22-03-2026	Replacement
M0004	EQ006	U0003	25-03-2026	Handle tightening	30000	25-03-2026	Repair
M0005	EQ003	U0005	25-03-2026	chain Oiling	90000	25-03-2026	Lubrication
M0006	EQ007	U0004	28-03-2026	Cultivation	1700000	02-03-2026	Cultivation

6 rows in set (0.00 sec)

FuelLog entity

mysql> SELECT * FROM fuelLog;

fuelId	equipmentId	date	quantity	cost	userName
FL001	EQ001	22/04/2026	10	66500	Mubbiru Clever
FL002	EQ003	20/04/2026	10	66500	Kwesiga Devine
FL003	EQ005	25/04/2026	4	31500	Kabunga Charles
FL004	EQ004	29/04/2026	20	140000	Tusiime Maria Desire

4 rows in set (0.00 sec)

Expense entity

mysql> select * from expense;

expenseId	equipmentId	expenseCategory	amount	date	userId
EX001	EQ001	Fuel	70000	20-03-2026	U0001
EX002	EQ002	Maintenance	150000	22-03-2026	U0002
EX003	EQ003	Lubrication	80000	25-03-2026	U0003
EX004	EQ004	Maintenance	150000	24-03-2026	U0004
EX004	EQ005	Maintenance	90000	28-03-2026	U0004
EX006	EQ006	Repair	140000	25-03-2026	U0006
EX007	EQ007	Fuel	30000	29-03-2026	U0007
EX008	EQ008	Spare Parts	75000	30-03-2026	U0008

8 rows in set (0.00 sec)

Equipment entity

mysql> SELECT * FROM equipment;;

equipmentId	equipmentName	categoryId	categoryName	status
EQ001	Lawn mower	CAT01	Garden Equipments	Needs a bigger grass collector
EQ002	Welding Machine	CAT02	Maintenance Tools	Works very well
EQ003	Chain Saw	CAT03	Garden Equipment	Needs a sharper blade
EQ004	Ox Plow	CAT04	Garden Equipments	Requires a stronger carrier
EQ005	Electric Drill	CAT05	Power Tools	Very efficient
EQ006	Hammer	CAT06	Hand Tools	Works well though not firm
EQ007	Tractor	CAT07	Garden Equipment	Works well but needs gears

7 rows in set (0.00 sec)

Deployment entity

mysql> SELECT * FROM deployment;

deploymentId	equipmentId	userId	startDate	endDate	taskName
DP004	EQ001	U0004	20/04/2026	25/04/2026	Weeding
DP002	EQ005	U0001	24/04/2026	28/04/2026	Maintenance
DP003	EQ005	U0002	23/04/2026	27/04/2026	Cultivation
DP001	EQ003	U0003	22/04/2026	23/04/2026	Spraying
DP005	EQ002	U0005	28/04/2026	4/05/2026	Repairing Tractor

5 rows in set (0.00 sec)

Branch entity

```
mysql> SELECT * FROM branch;
+----------+---------------------+-----------+
| branchId | manager             | branchName |
+----------+---------------------+-----------+
|        1 | Kabunga Charles     | Mbarara   |
|        2 | Tusiime Maria Desire | Kampala   |
|        3 | Kwesiga Devine      | Lukaya    |
|        4 | Nakalema Betty      | Masaka    |
|        5 | Mubiru Clever       | Kirumba   |
+----------+---------------------+-----------+
5 rows in set (0.01 sec)
```

QUERY COMMANDING USING DML ON THE LOGICAL SCHEMA

I realised i had forgotten to add a column in expense table and i used the **ALTER TABLE ‘ADD COLUMN‘** command to add userId as both tables have a relationship and userId will referebnce user table in the expense table helping in easy report making

```
mysql> ALTER TABLE expense ADD COLUMN userId VARCHAR(5);
Query OK, 0 rows affected (0.37 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

I used the **UPDATE** command to update record values misentered. Thus easy management due to data independency

```
mysql> UPDATE expense SET  userId="U0002" WHERE expenseId="EX002";
Query OK, 1 row affected (0.13 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE expense SET  userId="U0003" WHERE expenseId="EX003";
Query OK, 1 row affected (0.20 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE expense SET  userId="U0004" WHERE expenseId="EX004";
Query OK, 1 row affected (0.12 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE expense SET  userId="U0004" WHERE expenseId="EX005";
Query OK, 1 row affected (0.25 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

In some tables, i had not set the primary key and set them using **ALTER TABLE ... ‘ADD PRIMARY KEY‘ ..** command

```
mysql> ALTER TABLE fuelLog ADD PRIMARY KEY (fuelId);
Query OK, 0 rows affected (1.99 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE deployment ADD PRIMARY KEY (deploymentId);
Query OK, 0 rows affected (1.49 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE branch ADD PRIMARY KEY (branchId);
Query OK, 0 rows affected (1.97 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Selecting only users located in Mbarara easing task allocation

```
mysql> SELECT * FROM user WHERE location = "Mbarara";
+--------+----------------------+----------------------------+-----------+-------------------+----------+--------+------------+
| userId | name                 | email                      | phone     | qualification     | location | roleId | roleName   |
+--------+----------------------+----------------------------+-----------+-------------------+----------+--------+------------+
| U0002  | Tusiime Maria Desire | tusiimemariad001@gmail.com | 776332126 | BSc Agriculture   | Mbarara  | R003   | Superviser |
+--------+----------------------+----------------------------+-----------+-------------------+----------+--------+------------+
1 row in set (0.00 sec)
```

Selecting records in expense table whose amount is greater than 100000

This eases easy filtering of fields with a higher expense for better planning

```
mysql> SELECT * FROM expense WHERE amount > 100000;
+-----------+-------------+-----------------+--------+------------+--------+
| expenseId | equipmentId | expenseCategory | amount | date       | userId |
+-----------+-------------+-----------------+--------+------------+--------+
| EX002     | EQ002       | Maintenance     | 150000 | 22-03-2026 | U0002  |
| EX004     | EQ004       | Maintenance     | 150000 | 24-03-2026 | U0004  |
| EX006     | EQ006       | Repair          | 140000 | 25-03-2026 | U0006  |
+-----------+-------------+-----------------+--------+------------+--------+
3 rows in set (0.00 sec)
```

Using relational calculus **JOIN** command, i retrieved records in some attributes from **maintenance** and **user** tables with the same userId.

This enables quick reporrt making by knowing the operators and the respective tasks handled.

```
mysql> SELECT m.maintenanceId, u.name, m.description, m.amount, m.date FROM maintenance m JOIN user u ON m.userId = u.userId;
+---------------+----------------------+---------------------+---------+------------+
| maintenanceId | name                 | description         | amount  | date       |
+---------------+----------------------+---------------------+---------+------------+
| M0001         | Kwesiga Devine       | Welding rod replace |  150000 | 20-03-2026 |
| M0002         | Tusiime Maria Desire | Blade shappening    |  150000 | 24-03-2026 |
| M0004         | Tumwiine David       | Handle tightening   |   30000 | 25-03-2026 |
| M0005         | Kabunga Charles      | chain Oiling        |   90000 | 25-03-2026 |
| M0006         | Mubbiru Clever       | Cultivation         | 1700000 | 28-03-2026 |
+---------------+----------------------+---------------------+---------+------------+
5 rows in set (0.00 sec)
```

I ordered the records in maintenance table by date in ascending order, this helps a superviser to quickly know the sheduled tasks in their order and accomplished tasks for easy maintainability

```
mysql> SELECT * FROM maintenance ORDER BY date ASC;
+---------------+-------------+--------+------------+---------------------+---------+------------+-------------+
| maintenanceId | equipmentId | userId | date       | description         | amount  | endDate    | taskName    |
+---------------+-------------+--------+------------+---------------------+---------+------------+-------------+
| M0001         | EQ002       | U0001  | 20-03-2026 | Welding rod replace |  150000 | 21-03-2026 | Replacement |
| M0003         | EQ005       | EQ004  | 22-03-2026 | Drill bit change    |  110000 | 22-03-2026 | Replacement |
| M0002         | EQ004       | U0002  | 24-03-2026 | Blade shappening    |  150000 | 25-03-2026 | Sharpening  |
| M0004         | EQ006       | U0003  | 25-03-2026 | Handle tightening   |   30000 | 25-03-2026 | Repair      |
| M0005         | EQ003       | U0005  | 25-03-2026 | chain Oiling        |   90000 | 25-03-2026 | Lubrication |
| M0006         | EQ007       | U0004  | 28-03-2026 | Cultivation         | 1700000 | 02-03-2026 | Cultivation |
+---------------+-------------+--------+------------+---------------------+---------+------------+-------------+
6 rows in set (0.00 sec)
```

Conclusion

Farm Machinery and Equipment Management Database System provides a structured and the best efficient approach to managing agricultural machinery. By improving equipment tracking, maintenance scheduling, and operational planning, the system enhances productivity and reduces operational costs. Despite challenges such as user adoption and maintenance costs, the long-term benefits significantly justify its implementation, making it a viable and impactful solution in the agricultural sector.

Database Administrator and Designer: Kabunga Charles